CLASSIFICATION OF MATERIALS

In general, we may use the magnetic susceptibility $\chi_m$or the relative permeability μ_r to classify materials in terms of their magnetic property or behavior.

A material is said to be *nonmagnetic* if $\chi_m = 0\ (or\ \mu_r = 1)$; it is magnetic otherwise.

Free space, air, and materials with $\chi_m = 0\ (or\ \mu_r \approx 1$ are regarded as nonmagnetic.

Materials may be grouped into three major classes: diamagnetic, paramagnetic, and ferromagnetic as shown in fig 23below:

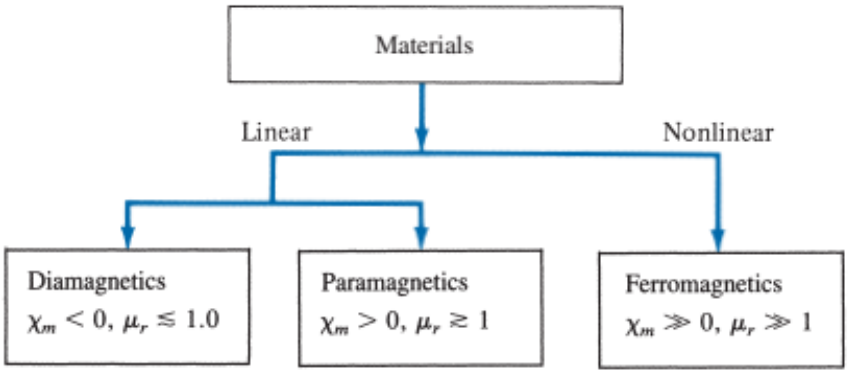

FIGURE 23 Classification of materials.

TABLE 5 Relative Permeability (μ_r) of Some Materials

Material	μ_r
Diamagnetic	
Bismuth	0.999833
Mercury	0.999968
Silver	0.9999736
Lead	0.9999831
Copper	0.9999906
Water	0.9999912
Hydrogen (STP)	$\simeq$ 1.0
Paramagnetic	
Oxygen (STP)	0.999998
Air	1.00000037
Aluminum	1.000021
Tungsten	1.00008
Platinum	1.0003
Manganese	1.001

Ferromagnetic	
Cobalt	250
Nickel	600
Soft iron	5000
Silicon–iron	7000

A material is said to be *diamagnetic* if it has $\mu_r \leq 1$ (i.e., very small negative χ_m)

It is *paramagnetic* if $\mu_r \geq 1$ (i.e., very small positiveχ_m.

From the table, it is apparent that for most practical purposes we may assume that $\mu_r \approx 1$ for diamagnetic and paramagnetic materials. Thus, we may regard diamagnetic and paramagnetic materials as linear and nonmagnetic.

If $\mu_r \gg 1$ (i.e., very large positive χ_m), the material is *ferromagnetic*. Ferromagnetic materials are always nonlinear and magnetic except when their temperatures are above curie temperature.

Diamagnetism occurs when the magnetic fields in a material that are due to electronic motions of orbiting and spinning completely cancel each other. Thus, the permanent (or intrinsic) magnetic moment of each atom is zero and such materials are weakly affected by a magnetic field. For most diamagnetic materials (e.g., bismuth, lead, copper, silicon, diamond, sodium chloride), χ_m is of the order of -10^{-5}. In certain materials, called *superconductors*, "perfect diamagnetism" occurs at temperatures near absolute zero:$\chi_m = -1 \; or \; \mu_r = 0 \; and \; B = 0$. Thus superconductors cannot contain magnetic field.

The diamagnetic properties of materials are seldom used in practice.

Paramagnetism occurs when the magnetic fields produced in a material by orbital and spinning electrons do not cancel completely. Unlike diamagnetism, paramagnetism is temperature dependent. For most paramagnetic materials (e.g., air, platinum, tungsten, potassium), χ_m is of the order $+10^{-5} \; to \; +10^{-3}$and is temperature dependent. Such materials find application in masers.

Ferromagnetism occurs in materials whose atoms have relatively large permanent magnetic moment. They are called ferromagnetic materials because the best-known member is iron. Other members are cobalt, nickel, and their alloys. Ferromagnetic materials are very useful in practice. As distinct from diamagnetic and paramagnetic materials, ferromagnetic materials have the following properties:

1. They are capable of being magnetized very strongly by a magnetic field.

2. They retain a considerable amount of their magnetization when removed from the field.

3. They lose their ferromagnetic properties and become linear paramagnetic materials when the temperature is raised above a certain temperature known as the *curie temperature.* Thus if a permanent magnet is heated above its curie temperature

(770°C for iron), it loses its magnetization completely.

4. They are nonlinear; that is, the constitutive relation $B = \mu_0 \mu_r H$ does not hold for ferromagnetic materials because μ_r depends on **B** and cannot be represented by a single value.

Even though $B = \mu_0(H + M)$ holds for all materials including ferromagnetic, the relationship between **B** and **H** depends on previous magnetization of a ferromagnetic material—its "magnetic history." Instead of having a linear relationship between **B** and **H** (i.e., $\boldsymbol{B} = \boldsymbol{\mu H}$), it is only possible to represent the relationship by a *magnetization curve* or *B–H* curve.

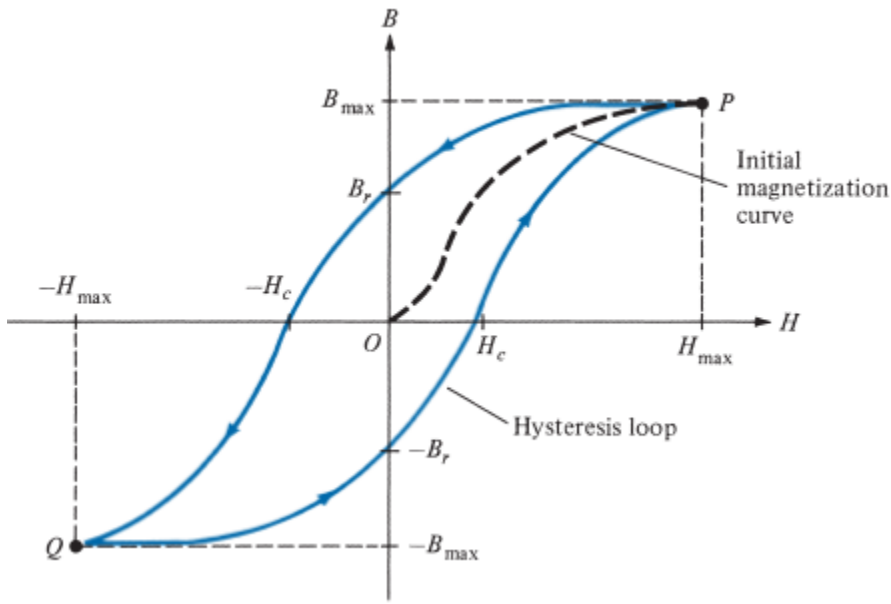

FIGURE 24 Typical magnetization (*B–H*) curve.

A typical *B–H* curve is shown in Figure 24. First, note the nonlinear relationship between *B* and *H*. Second, at any point on the curve μ is given by the ratio *B/H* and not by *dB/dH*, the slope of the curve.

If we assume that the ferromagnetic material whose *B–H* curve in Figure is initially unmagnetized, as *H* increases (owing to increase in current) from *O* to maximum applied field intensity *H*max, curve *OP* is produced. This curve is referred to as the *virgin* or *initial magnetization curve.* After reaching saturation at *P*, if *H* is decreased, *B* does not follow the initial curve but lags behind *H*. This phenomenon of *B* lagging behind *H* is called *hysteresis*.

If *H* is reduced to zero, *B* is not reduced to zero but to *Br*, which is referred to as the *permanent flux density*. The value of *Br* depends on *H*max, the maximum applied field intensity. The existence of *Br* is the cause of having permanent magnets. If *H* increases negatively (by reversing the direction of current), *B* becomes zero when *H* becomes *Hc*, which is known as the *coercive field intensity*. Materials for which *Hc* is small are said to be magnetically hard. The value of *Hc* also depends on *H*max.

Further increase in H in the negative direction to reach Q and a reverse in its direction to reach P gives a closed curve called a *hysteresis loop.*

Hysteresis loops vary in shape from one material to another. Some ferrites, for example, have an almost rectangular hysteresis loop and are used in digital computers as magnetic information storage devices.

The area of a hysteresis loop gives the energy loss (hysteresis loss) per unit volume during one cycle of the periodic magnetization of the ferromagnetic material. This energy loss is in the form of heat. It is therefore desirable that materials used in electric generators, motors, and transformers have tall but narrow hysteresis loops so that hysteresis losses are minimal.

Example;

Region $0 \leq z \leq 2\ m$ is occupied by an infinite slab of permeable material ($\mu_r = 2.5$).

If $B = 10ya_x - 5xa_y\ \ y$ mWb/m^2 within the slab, determine:

(a) **J**,

(b) **J***b*,

(c) **M**,

(d) $\mathbf{K}_b$ on z= 0.

Solution:

(a) By definition;

$$\mathbf{J} = \nabla \times \mathbf{H} = \nabla \times \frac{\mathbf{B}}{\mu_o \mu_r} = \frac{1}{4\pi \times 10^{-7}(2.5)}\left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y}\right)\mathbf{a}_z$$

$$= \frac{10^6}{\pi}(-5 - 10)10^{-3}\mathbf{a}_z = -4.775\mathbf{a}_z \text{ kA/m}^2$$

(b) $$\mathbf{J}_b = \chi_m \mathbf{J} = (\mu_r - 1)\mathbf{J} = 1.5(-4.775\mathbf{a}_z) \cdot 10^3$$
$$= -7.163\mathbf{a}_z \text{ kA/m}^2$$

(c) $$\mathbf{M} = \chi_m \mathbf{H} = \chi_m \frac{\mathbf{B}}{\mu_o \mu_r} = \frac{1.5(10y\mathbf{a}_x - 5x\mathbf{a}_y) \cdot 10^{-3}}{4\pi \times 10^{-7}(2.5)}$$
$$= 4.775y\mathbf{a}_x - 2.387x\mathbf{a}_y \text{ kA/m}$$

(d) $\mathbf{K}_b = \mathbf{M} \times \mathbf{a}_n$. Since $z = 0$ is the lower side of the slab occupying $0 \leq z \leq 2$, $\mathbf{a}_n = -\mathbf{a}_z$. Hence,

$$\mathbf{K}_b = (4.775y\mathbf{a}_x - 2.387x\mathbf{a}_y) \times (-\mathbf{a}_z)$$
$$= 2.387x\mathbf{a}_x + 4.775y\mathbf{a}_y \text{ kA/m}$$

MAGNETIC BOUNDARY CONDITIONS

We define magnetic boundary conditions as the conditions that H (or B) field must satisfy at the boundary between two different media.

We make use of Gauss's law for magnetic fields

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0 \qquad 1$$

and Ampère's circuit law

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = I \qquad 2$$

Consider the boundary between two magnetic media 1 and 2, characterized, respectively, by μ_1 and μ_2 as in Figure 25.

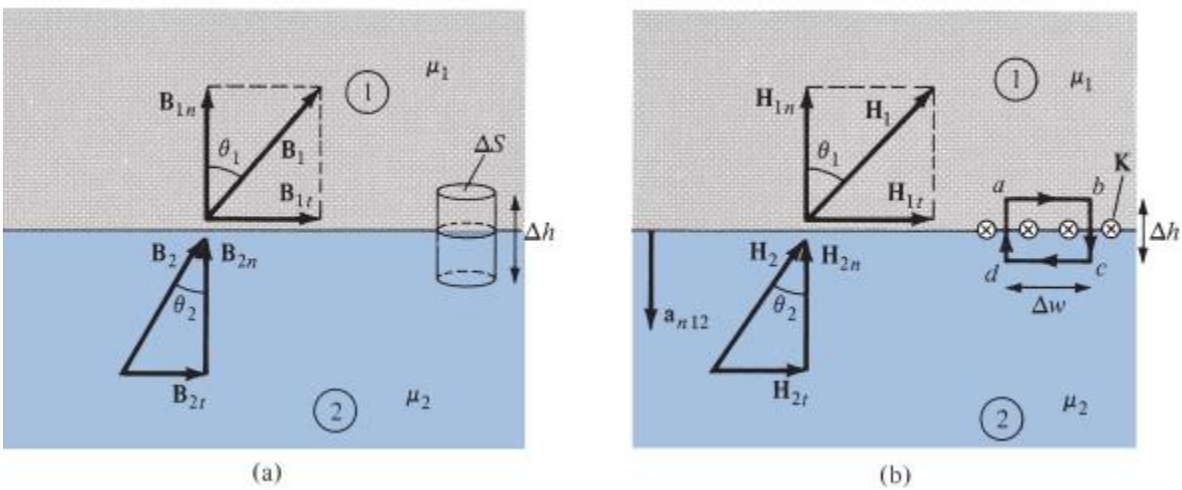

FIGURE 25 Boundary conditions between two magnetic media: (**a**) for **B**, (**b**) for **H**.

Applying eq. (1) to the pillbox (Gaussian surface) of Figure 25(a) and allowing $\Delta h \to 0$, we obtain

$$B_{1n}\,\Delta S - B_{2n}\,\Delta S = 0 \qquad 3$$

Thus

$$\boxed{\mathbf{B}_{1n} = \mathbf{B}_{2n}} \quad \text{or} \quad \mu_1 \mathbf{H}_{1n} = \mu_2 \mathbf{H}_{2n} \qquad 4$$

Since $B = \mu H$. Equation (4) shows that the normal component of **B** is continuous at the boundary. It also shows that the normal component of **H** is discontinuous at the boundary; **H** undergoes some change at the interface.

Similarly, we apply eq. (2) to the closed path *abcda* of Figure 25(b), where surface current **K** on the boundary is assumed normal to the path. We obtain;

$$K \cdot \Delta w = H_{1t} \cdot \Delta w + H_{1n} \cdot \frac{\Delta h}{2} + H_{2n} \cdot \frac{\Delta h}{2} \qquad 5$$

$$-H_{2t} \cdot \Delta w - H_{2n} \cdot \frac{\Delta h}{2} - H_{1n} \cdot \frac{\Delta h}{2} \qquad 6$$

As $\Delta h \to 0$, eq. (6) leads to

$$H_{1t} - H_{2t} = K \qquad 7$$

This shows that the tangential component of *H* is also discontinuous. Equation (7) may be written in terms of *B* as

$$\frac{B_{1t}}{\mu_1} - \frac{B_{2t}}{\mu_2} = K \qquad 8$$

In the general case, eq. (7) becomes

$$(\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_{n12} = \mathbf{K} \qquad 9$$

Where a_{n12} is a unit vector normal to the interface and is directed from medium 1 to medium 2. If the boundary is free of current or the media are not conductors (for **K** is free current density), **K** = **0** and eq. (7) becomes;

$$\boxed{\mathbf{H}_{1t} = \mathbf{H}_{2t}} \quad \text{or} \quad \frac{\mathbf{B}_{1t}}{\mu_1} = \frac{\mathbf{B}_{2t}}{\mu_2} \qquad 10$$

Thus the tangential component of H is continuous while that of B is discontinuous at the boundary.

If the fields make an angle θ with the normal to the interface, eq. (4) results in

$$B_1 \cos\theta_1 = B_{1n} = B_{2n} = B_2 \cos\theta_2 \qquad 11$$

while eq. (10) produces

$$\frac{B_1}{\mu_1} \sin\theta_1 = H_{1t} = H_{2t} = \frac{B_2}{\mu_2} \sin\theta_2 \qquad 12$$

Dividing eq. (12) by eq. (11) gives

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2} \qquad 13$$

INDUCTORS AND INDUCTANCES

A circuit (or closed conducting path) carrying current I produces a magnetic field **B** that causes a flux $\psi = \int Bds$ to pass through each turn of the circuit as shown in Figure 26.

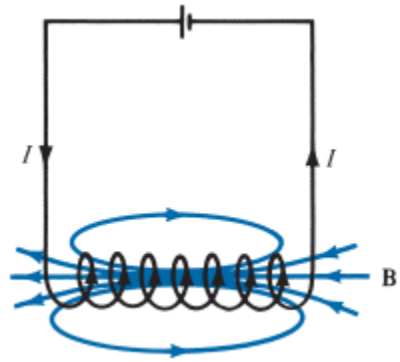

FIGURE 26 Magnetic field **B** produced by a circuit

If the circuit has N identical turns, we define the *flux linkage* λ as;

$$\lambda = N\psi \qquad 1$$

Also, if the medium surrounding the circuit is linear, the flux linkage λ is proportional to the current I producing it; that is, $\lambda \propto I$ or

$$\lambda = LI \qquad 2$$

where L is a constant of proportionality called the *inductance* of the circuit. The inductance L is a property of the physical arrangement of the circuit. It is the ability of the physical arrangement to store magnetic energy.

A circuit or part of a circuit that has inductance is called an *inductor*. The **inductance** L of an inductor is the ratio of the magnetic flux linkage λ to the current I through the inductor.

$$L = \frac{\lambda}{I} = \frac{N\Psi}{I} \qquad 3$$

The unit of inductance is the henry (H), which is the same as webers per ampere. Since the henry is a fairly large unit, inductances are usually expressed in millihenrys (mH).

The inductance defined by eq. (3) is commonly referred to as *self-inductance*, since the linkages are produced by the inductor itself. Like capacitance, inductance may be regarded as a measure of how much magnetic energy is stored in an inductor. The magnetic energy (in joules) stored in an inductor is expressed in circuit theory as;

$$W_m = \frac{1}{2}LI^2 \qquad 4$$

or

$$L = \frac{2W_m}{I^2} \qquad 5$$

Thus the self-inductance of a circuit may be defined or calculated from energy considerations.

Mutual inductance

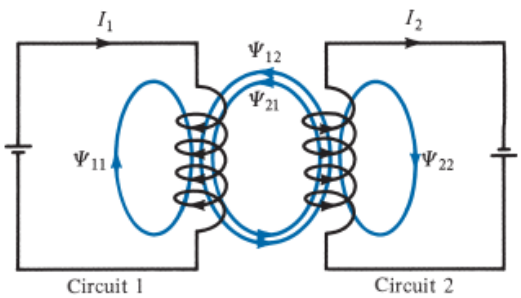

FIGURE 27 Magnetic interaction between two circuits.

If instead of having a single circuit, we have two circuits carrying current I_1 and I_2 as shown in Figure 27, a magnetic interaction exists between the circuits. Four component fluxes ψ_{11}, ψ_{12} , ψ_{21}, and ψ_{22} are produced. The flux ψ_{12}, for example, is the flux passing through circuit 1 due to current I_2 in circuit 2. If $\mathbf{B}_2$ is the magnetic flux density due to I_2 and S_1 is the area of circuit 1, then

$$\Psi_{12} = \int_{S_1} \mathbf{B}_2 \cdot d\mathbf{S} \qquad 6$$

The *mutual inductance* M_{12} is the ratio of the flux linkage $\lambda_{12} = N_1\psi_{12}$ on circuit 1 to current I_2.

$$M_{12} = \frac{\lambda_{12}}{I_2} = \frac{N_1\Psi_{12}}{I_2} \qquad 7$$

Similarly, the mutual inductance M_{21} is defined as the flux linkages of circuit 2 per unit current I_1; that is,

$$M_{21} = \frac{\lambda_{21}}{I_1} = \frac{N_2\Psi_{21}}{I_1} \qquad 8$$

It can be shown by using energy concepts that if the medium surrounding the circuits is linear (i.e., in the absence of ferromagnetic material),

$$M_{12} = M_{21}$$

The mutual inductance M_{12} or M_{21} is expressed in henrys and should not be confused with the magnetization vector **M** expressed in amperes per meter. Mutual inductance is fundamental to the operation of transformers. We define the self-inductance of circuits 1 and 2, respectively, as

$$L_1 = \frac{\lambda_{11}}{I_1} = \frac{N_1 \Psi_1}{I_1}$$

And

$$L_2 = \frac{\lambda_{22}}{I_2} = \frac{N_2 \Psi_2}{I_2}$$

Where $\psi_1 = \psi_{11} + \psi_{12}$ and $\psi_2 = \psi_{21} + \psi_{22}$. The total energy in the magnetic field is the sum of the energies due to L_1, L_2, and M_{12} (or M_{21}); that is,

$$\begin{aligned} W_m &= W_1 + W_2 + W_{12} \\ &= \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 \pm M_{12} I_1 I_2 \end{aligned}$$

The positive sign is taken if currents I_1 and I_2 flow such that the magnetic fields of the two circuits strengthen each other. If the currents flow such that their magnetic fields oppose each other, the negative sign is taken.

Typical examples of inductors are toroids, solenoids, coaxial transmission lines, and parallel-wire transmission lines.

The inductance of each of these inductors can be determined by following a procedure;

1. 1.Choose a suitable coordinate system
2. Let the inductor carry current *I*.
3. Determine **B** from Biot–Savart's law (or from Ampère's law if symmetry exists) and calculate ψ from$\psi = \int B.ds$.
4. Finally find L from $L = \frac{\lambda}{I} = \frac{N\psi}{I}$